

CODE :-

```
import numpy as np
import numpy.random as ar
def inpt():
    b=[]
    for i in range (10):
        asc=ord(str(i))
        bin=format(asc,'07b')
        print(i,"->",asc,"->",bin)
        d=[]
        d=[int(i) for i in bin]
        b.append(d)
    #print(b)
    return b

x=inpt()
print("X=",x)
w=ar.rand(7)
print("W=",w)
t=[1,0,1,0,1,0,1,0,1,0]
b=0
alpha=0.1

for epoach in range(50):
    print("epoach=",epoach)
    e=0
    for i in range(len(x)):
        netw=np.dot(w,x[i])+b
        yin=1 if netw>=0 else 0
        error=t[i]-yin
        print("netw=",netw,"yin=",yin,"error=",error)
        if error!=0:
            e=e+1
            w=w+(alpha*error*np.array(x[i]))
            b=b+(alpha*error)
    if e==0:
        break
print("final W=",w)
print("final b=",b)

while True:
    a = int(input("Enter a number: "))
    asc = ord(str(a))
    bin_str = format(asc,'07b')
    d = [int(bit) for bit in bin_str]
    netw = np.dot(w,d) + b
    yin = 1 if netw >= 0 else 0
    print("Result:", "Even" if yin==1 else "Odd")
    choice = input("Do you want to continue? (y/n): ")
    if choice != 'y' and choice != 'Y':
        Break
```

OUTPUT :-

0 -> 48 -> 0110000

1 -> 49 -> 0110001

2 -> 50 -> 0110010

3 -> 51 -> 0110011

4 -> 52 -> 0110100

5 -> 53 -> 0110101

6 -> 54 -> 0110110

7 -> 55 -> 0110111

8 -> 56 -> 0111000

9 -> 57 -> 0111001

X= [[0, 1, 1, 0, 0, 0, 0], [0, 1, 1, 0, 0, 0, 1], [0, 1, 1, 0, 0, 1, 0], [0, 1, 1, 0, 0, 1, 1], [0, 1, 1, 0, 1, 0, 0], [0, 1, 1, 0, 1, 0, 1], [0, 1, 1, 0, 1, 1, 0], [0, 1, 1, 0, 1, 1, 1], [0, 1, 1, 1, 0, 0, 0], [0, 1, 1, 1, 0, 0, 1]]

W= [0.87867006 0.33011316 0.7536151 0.44744078 0.30227945 0.96415996
0.96792033]

epoch= 0

netw= 1.0837282625907827 yin= 1 error= 0

netw= 2.051648595697264 yin= 1 error= -1

netw= 1.7478882195492509 yin= 1 error= 0

netw= 2.615808552655732 yin= 1 error= -1

netw= 0.7860077098773108 yin= 1 error= 0

netw= 1.5539280429837923 yin= 1 error= -1

netw= 1.250167666835779 yin= 1 error= 0

netw= 1.9180879999422602 yin= 1 error= -1

netw= 0.3311690430788512 yin= 1 error= 0

netw= 0.8990893761853326 yin= 1 error= -1

epoch= 1

netw= -0.4162717374092173 yin= 0 error= 1

netw= 0.35164859569726403 yin= 1 error= -1

netw= 0.3478882195492512 yin= 1 error= 0

netw= 0.7158085526557327 yin= 1 error= -1

netw= -0.6139922901226892 yin= 0 error= 1

netw= 0.05392804298379228 yin= 1 error= -1

netw= 0.050167666835779334 yin= 1 error= 0

netw= 0.2180879999422608 yin= 1 error= -1

netw= -0.6688309569211488 yin= 0 error= 1

netw= -0.2009106238146674 yin= 0 error= 0

epoch= 2

netw= -0.7162717374092173 yin= 0 error= 1

netw= -0.3483514043027359 yin= 0 error= 0

netw= 0.14788821954925113 yin= 1 error= 0

netw= 0.2158085526557325 yin= 1 error= -1

netw= -0.7139922901226893 yin= 0 error= 1
netw= -0.34607195701620785 yin= 0 error= 0
netw= 0.1501676668357792 yin= 1 error= 0
netw= 0.11808799994226071 yin= 1 error= -1
netw= -0.2688309569211488 yin= 0 error= 1
netw= -0.0009106238146674439 yin= 0 error= 0

epoch= 3

netw= -0.4162717374092174 yin= 0 error= 1
netw= -0.248351404302736 yin= 0 error= 0
netw= 0.24788821954925122 yin= 1 error= 0
netw= 0.11580855265573253 yin= 1 error= -1
netw= -0.4139922901226893 yin= 0 error= 1
netw= -0.2460719570162079 yin= 0 error= 0
netw= 0.2501676668357793 yin= 1 error= 0
netw= 0.018087999942260735 yin= 1 error= -1
netw= 0.13116904307885113 yin= 1 error= 0
netw= -0.2009106238146674 yin= 0 error= 0

epoch= 4

netw= -0.4162717374092174 yin= 0 error= 1
netw= -0.448351404302736 yin= 0 error= 0
netw= 0.047888219549251154 yin= 1 error= 0
netw= -0.2841914473442674 yin= 0 error= 0
netw= -0.1139922901226893 yin= 0 error= 1
netw= -0.04607195701620792 yin= 0 error= 0
netw= 0.45016766683577925 yin= 1 error= 0
netw= 0.11808799994226066 yin= 1 error= -1
netw= 0.4311690430788512 yin= 1 error= 0
netw= -0.0009106238146674994 yin= 0 error= 0

epoch= 5

netw= -0.11627173740921742 yin= 0 error= 1
netw= -0.24835140430273608 yin= 0 error= 0
netw= 0.2478882195492511 yin= 1 error= 0
netw= -0.18419144734426748 yin= 0 error= 0
netw= 0.18600770987731063 yin= 1 error= 0
netw= -0.246071957016208 yin= 0 error= 0
netw= 0.2501676668357793 yin= 1 error= 0
netw= -0.1819120000577394 yin= 0 error= 0
netw= 0.7311690430788509 yin= 1 error= 0
netw= 0.2990893761853325 yin= 1 error= -1

epoch= 6

netw= -0.11627173740921742 yin= 0 error= 1
netw= -0.34835140430273603 yin= 0 error= 0

netw= 0.2478882195492511 yin= 1 error= 0
netw= -0.2841914473442675 yin= 0 error= 0
netw= 0.18600770987731063 yin= 1 error= 0
netw= -0.34607195701620797 yin= 0 error= 0
netw= 0.2501676668357793 yin= 1 error= 0
netw= -0.28191200005773936 yin= 0 error= 0
netw= 0.6311690430788511 yin= 1 error= 0
netw= 0.09908937618533253 yin= 1 error= -1

epoch= 7

netw= -0.11627173740921742 yin= 0 error= 1
netw= -0.448351404302736 yin= 0 error= 0
netw= 0.2478882195492511 yin= 1 error= 0
netw= -0.38419144734426747 yin= 0 error= 0
netw= 0.18600770987731063 yin= 1 error= 0
netw= -0.44607195701620794 yin= 0 error= 0
netw= 0.2501676668357793 yin= 1 error= 0
netw= -0.38191200005773934 yin= 0 error= 0
netw= 0.5311690430788512 yin= 1 error= 0
netw= -0.10091062381466745 yin= 0 error= 0

epoch= 8

netw= 0.18372826259078257 yin= 1 error= 0
netw= -0.448351404302736 yin= 0 error= 0
netw= 0.2478882195492511 yin= 1 error= 0
netw= -0.38419144734426747 yin= 0 error= 0
netw= 0.18600770987731063 yin= 1 error= 0
netw= -0.44607195701620794 yin= 0 error= 0
netw= 0.2501676668357793 yin= 1 error= 0
netw= -0.38191200005773934 yin= 0 error= 0
netw= 0.5311690430788512 yin= 1 error= 0
netw= -0.10091062381466745 yin= 0 error= 0
final W= [0.87867006 0.03011316 0.4536151 0.34744078 0.00227945 0.06415996
-0.63207967]
final b= -0.30000000000000004

Enter a number: 9

Result: Odd

Do you want to continue? (y/n): y

Enter a number: 6

Result: Even

Do you want to continue? (y/n): n